BRAC UNIVERSITY
Inspiring Excellence

Linear Algebra & Fourier Analysis Assessment

December 23, 2025

Total - 20 Marks

Name:

ID:

Section:

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1 Sets

1. (a) **C12: (Ellipse locus)** [5 Marks]

Identify and sketch the locus defined by

$$|z-1|+|z+1|=4.$$

- State the foci.
- Find the length of the major axis.

(b) **C7: (Fourth roots)** [5 Marks]

Find all fourth roots of

$$z=16i.$$

Write your answers in:

- Polar form
- Rectangular form

(c) **C5: (De Moivre)** [5 Marks]

Compute, in simplest rectangular form,

$$(1-i)^8.$$

(d) [5 Marks]

Sketch step-by-step the graph of

$$y = 1 + 2\cos\left(3x - \frac{\pi}{2}\right).$$

- Clearly indicate each transformation.
- Mark at least one full period.

Best of Luck!